

Take Home Assignment 02

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August 2026

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Question 01

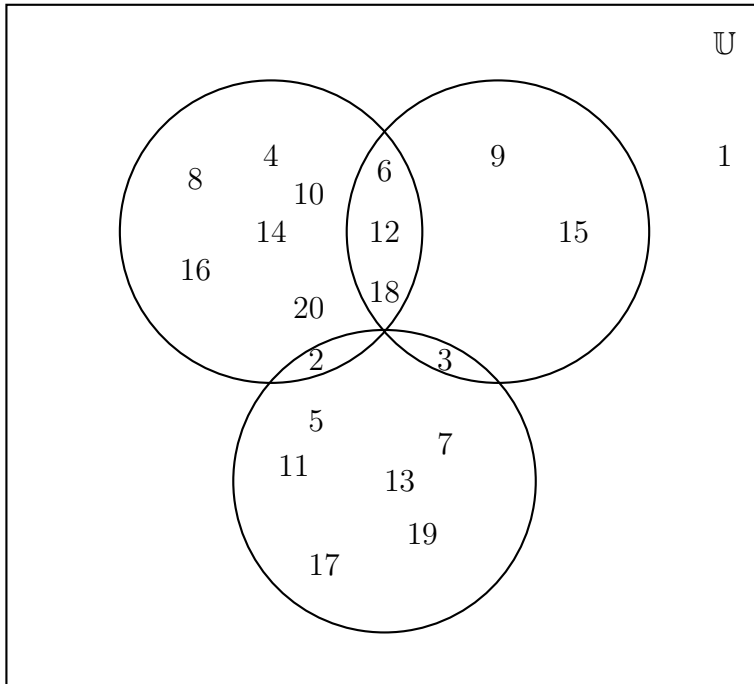
(a)

$$\mathbb{U} = \{1, 2, 3, 4, \dots, 20\}$$

$$A = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$$

$$B = \{3, 6, 9, 12, 15, 18\}$$

$$C = \{2, 3, 5, 7, 11, 13, 17, 19\}$$



From the above Venn Diagram :

(i) $(\mathbf{A} \cup \mathbf{B})^c = \{1, 5, 7, 11, 13, 17, 19\}$

(ii) $(\mathbf{A} \cap \mathbf{B}) \cup \mathbf{C} = \{2, 3, 5, 6, 7, 11, 12, 13, 17, 18, 19\}$

(iii) $(\mathbf{A} - \mathbf{C}) \Delta (\mathbf{B} - \mathbf{C})$

$$(\mathbf{A} - \mathbf{C}) = \{4, 6, 8, 10, 12, 14, 16, 18, 20\}$$

$$(\mathbf{B} - \mathbf{C}) = \{6, 9, 12, 15, 18\}$$

$$(\mathbf{A} - \mathbf{C}) \Delta (\mathbf{B} - \mathbf{C}) = \{4, 8, 9, 10, 14, 15, 16, 20\}$$

(iv) $((\mathbf{A} \cup \mathbf{C}) \cap (\mathbf{B} \cup \mathbf{C})) - \mathbf{C}$

$$(\mathbf{A} \cup \mathbf{C}) = \{2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20\}$$

$$(\mathbf{B} \cup \mathbf{C}) = \{2, 3, 5, 6, 7, 9, 11, 12, 13, 15, 17, 18, 19\}$$

$$(\mathbf{A} \cup \mathbf{C}) \cap (\mathbf{B} \cup \mathbf{C}) = \{2, 3, 5, 6, 7, 11, 12, 13, 17, 18, 19\}$$

$$((\mathbf{A} \cup \mathbf{C}) \cap (\mathbf{B} \cup \mathbf{C})) - \mathbf{C} = \{6, 12, 18\}$$

$$(v) (A \cup C) \cap (B \cup C) \equiv (A \cap B) \cup C$$

L.H.S. = $\{2, 3, 5, 6, 7, 11, 12, 13, 17, 18, 19\}$ (As written above)

R.H.S. = $\{2, 3, 5, 6, 7, 11, 12, 13, 17, 18, 19\}$ (As written above)

Hence $(A \cup C) \cap (B \cup C) \equiv (A \cap B) \cup C$

(b)

$$X = \{a, b, c, d, e\}$$

Cardinality of x ($|X|$) is 5.

(i) Contain a :

$$2^{5-1} = 2^4 = 16$$

Counting argument : a is kept fixed, and other 4 elements can either include or exclude in the subset.

(ii) Contain both a and b :

$$2^{5-2} = 2^3 = 8$$

Counting argument : a and b is kept fixed, and other 3 elements can either include or exclude in the subset.

(iii) Contain exactly three items :

$${}^5C_3 = \frac{5!}{3! \cdot 2!} = \frac{5 \cdot 4}{2} = 10$$

(iv) Counting argument : Since the subset is made using any 3 elements out of 5, total amount of can be calculated using combinatorics since the order doesn't matter.

(v) Contain a but not e :

$$2^{5-2} = 2^3 = 8$$

Counting argument : a is kept fixed and e is excluded, and other 3 elements can either include or exclude in the subset.

(c)

$$A - (B \cup C) = (A - B) \cap (A - C)$$

$$A - (B \cup C) \subseteq (A - B) \cap (A - C)$$

$$\text{Let } x \in A - (B \cup C)$$

$$x \in A \text{ and } x \notin (B \cup C) \text{ By definition of set difference}$$

$$x \in A \text{ and } x \notin B \text{ and } x \notin C \text{ By De Morgan's Law on set operations}$$

$$x \in A - B \text{ and } x \in A - C \text{ By definition of set difference}$$

$$x \in ((A - B) \cap (A - C))$$

$$\therefore A - (B \cup C) \subseteq (A - B) \cap (A - C)$$

$$(A - B) \cap (A - C) \subseteq A - (B \cup C)$$

$$\text{Let } x \in ((A - B) \cap (A - C))$$

$$x \in A - B \text{ and } x \in A - C$$

$$x \in A \text{ and } x \notin B \text{ and } x \notin C \text{ By definition of set difference}$$

$$x \in A \text{ and } x \notin (B \cup C) \text{ By De Morgan's Law on set operations}$$

$$x \in A - (B \cup C) \text{ By definition of set difference}$$

$$\therefore (A - B) \cap (A - C) \subseteq A - (B \cup C)$$

$$\therefore A - (B \cup C) = (A - B) \cap (A - C)$$

(d)

$$\mathbb{U} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$A = 101101001011$$

$$B = 011011100101$$

$$C = 110010011010$$

$$(i) (A \cap B^c) \cup C$$

$$B^c = \text{NOT}(011011100101) = 100100011010$$

$$A \cap B^c = 101101001011 \text{ Bitwise AND } 100100011010 = 100100001010$$

$$(A \cap B^c) \cup C = 100100001010 \text{ Bitwise OR } 110010011010 = 110110011010$$

$$(A \cap B^c) \cup C = \{1, 2, 4, 5, 8, 9, 11\}$$

$$(ii) (A \Delta B) \cap C^c$$

$$C^c = \text{NOT}(110010011010) = 001101100101$$

$$A \Delta B = 101101001011 \text{ Bitwise XOR } 011011100101 = 110110101110$$

$$(A \Delta B) \cap C^c = 110110101110 \text{ Bitwise AND } 001101100101 = 000100100100$$

$$(A \Delta B) \cap C^c = \{4, 7, 10\}$$

(e)

$$[(A \cup B) \cap (A \cup B^c)] \cup [A^c \cap (B \cup B^c)]$$

$$[(A \cap A) \cup (A \cap B^c) \cup (A \cap B) \cup (B \cap B^c)] \cup [A^c \cap (B \cup B^c)] \quad [\text{Distributive law for first bracket}]$$

$$[A \cup (A \cap B^c) \cup (A \cap B) \cup (B \cap B^c)] \cup [A^c \cap (B \cup B^c)] \quad [\text{Idempotent law}]$$

$$[A \cup (A \cap B^c) \cup (A \cap B) \cup \emptyset] \cup [A^c \cap \mathbb{U}] \quad [\text{Complement law}]$$

$$[A \cup (A \cap B^c) \cup (A \cap B)] \cup A^c \quad [\text{Identity law}]$$

$$A \cup A^c \quad [\text{Redundancy law}]$$

$$\mathbb{U} \quad [\text{Complement law}]$$

Let $x \in \mathbb{U}$

$$x \in B \text{ or } x \notin B (x \in B^c)$$

$$\therefore x \in (B \cup B^c)$$

Similarly $x \in A \text{ or } x \notin A (x \in A^c)$

For $x \in A$:

$$x \in (A \cup B) \text{ and } x \in (A \cup B^c)$$

$$x \in ((A \cup B) \cap (A \cup B^c)) \quad [\text{By definition of intersection}]$$

For $x \in A^c$:

Since $x \in A^c$ and $x \in (B \cup B^c)$

$$x \in (A^c \cap (B \cup B^c)) \quad [\text{By definition of intersection}]$$

$$\therefore x \in ((A \cup B) \cap (A \cup B^c)) \cup (A^c \cap (B \cup B^c)) \quad [\text{By definition of union}]$$

$$\therefore [(A \cup B) \cap (A \cup B^c)] \cup [A^c \cap (B \cup B^c)] = \mathbb{U}$$

Question 02

(a)

Domain : $A = \{-2, -1, 0, 1, 2\}$

Codomain : $\mathbb{Z}$ (set of integers)

Function : $f(x) = x^2 - x$

(i) Range of f

$$f(-2) = (-2)^2 - (-2) = 6$$

$$f(-1) = (-1)^2 - (-1) = 2$$

$$f(0) = (0)^2 - (0) = 0$$

$$f(1) = (1)^2 - (1) = 0$$

$$f(2) = (2)^2 - (2) = 2$$

Range of f = $\{0, 2, 6\}$

(ii) Here $f(0) = 0$ and $f(1) = 0$ also $f(-1) = 2$ and $f(2) = 2$

Since there are multiple elements in the domain of f which maps to the same element in range of f, f is not an one-to-one function.

(iii) $B = f(A) = \{0, 2, 6\}$

$$f : A \rightarrow B$$

New codomain : $\{0, 2, 6\}$

$$f(0) = 0 \text{ and } f(1) = 0$$

$$f(-1) = 2 \text{ and } f(2) = 2$$

$$f(-2) = 6$$

Since for every element of codomain of f, there exists at least one element of the domain of f which maps to it, the new function f is onto.

Still there are multiple elements in the domain of f which maps to the same element in range of f, new function f is not an one-to-one function.

$\therefore$ new function f is not bijective.

(b)Domain : All $\mathbb{R}$ except -1Codomain : All $\mathbb{R}$ except 2

$$\text{Function : } f = \frac{2x - 3}{x + 1}$$

- (i) For a function to be one-to-one
- $\forall a \forall b (f(a) = f(b) \rightarrow a = b)$
- in domain
- $\mathbb{R} \setminus \{-1\}$

$$\frac{2a - 3}{a + 1} = \frac{2b - 3}{b + 1}$$

$$(2a - 3)(b + 1) = (2b - 3)(a + 1)$$

$$(2ab + 2a - 3b - 3) = (2ab + 2b - 3a - 3)$$

$$2a - 3b = 2b - 3a$$

$$5a = 5b$$

$$a = b$$

Since $f(a) = f(b)$ implies $a = b$ then the function f is an one-to-one function.

- (ii) For a function to be onto
- $\forall y \exists x (f(x) = y)$
- x
- in domain and
- y
- in codomain.

$$y = \frac{2x - 3}{x + 1}$$

$$y(x + 1) = 2x - 3$$

$$xy + y = 2x - 3$$

$$2x - xy = y + 3$$

$$x(2 - y) = y + 3$$

$$x = \frac{y + 3}{2 - y} \text{ [y is in codomain which is } \mathbb{R} \setminus \{2\}]$$

Let $x = -1$

$$-1 = \frac{y + 3}{2 - y}$$

$$y - 2 = y + 3$$

$$-2 = 3 \text{ [contradiction for } x = -1]$$

Since $f(x) = y$ holds for domain and codomain of function f , the function f is an onto function.

(iii) Since f is both one-to-one and onto (Bijection) the function f has an inverse function.

$$f = \frac{2x - 3}{x + 1}$$

$$x = \frac{2f^{-1} - 3}{f^{-1} + 1}$$

$$x(f^{-1} + 1) = (2f^{-1} - 3)$$

$$xf^{-1} + x = (2f^{-1} - 3)$$

$$xf^{-1} - 2f^{-1} = -3 - x$$

$$f^{-1}(x - 2) = -(x + 3)$$

$$f^{-1} = \frac{-(x + 3)}{x - 2}$$

(iv) $f(f^{-1}(y)) = y$

$$\text{L.H.S.} = \frac{2f^{-1}(y) - 3}{f^{-1}(y) + 1}$$

$$\text{L.H.S.} = \frac{2\left(\frac{-(y + 3)}{y - 2}\right) - 3}{\left(\frac{-(y + 3)}{y - 2}\right) + 1}$$

$$\text{L.H.S.} = \frac{\frac{-2(y + 3) - 3(y - 2)}{y - 2}}{\frac{-(y + 3) + (y - 2)}{y - 2}}$$

$$\text{L.H.S.} = \frac{-2(y + 3) - 3(y - 2)}{-(y + 3) + (y - 2)}$$

$$\text{L.H.S.} = \frac{-2y - 6 - 3y + 6}{-y - 3 + y - 2}$$

$$\text{L.H.S.} = \frac{-5y}{-5}$$

$$\text{L.H.S.} = y = \text{R.H.S.}$$

(c)

$$f(x) = 2x - 1$$

$$g(x) = x^2 + 2$$

$$(i) \quad (f \circ g)(x) = f(g(x))$$

$$f(x^2 + 2)$$

$$2(x^2 + 2) - 1$$

$$2x^2 + 4 - 1$$

$$2x^2 + 3$$

$$(ii) \quad (g \circ f)(x) = g(f(x))$$

$$g(2x - 1)$$

$$(2x - 1)^2 + 2$$

$$4x^2 - 4x + 1 + 2$$

$$4x^2 - 4x + 3$$

$$(iii) \quad (f \circ f)(x) = f(f(x))$$

$$f(2x - 1)$$

$$2(2x - 1) - 1$$

$$4x - 2 - 1$$

$$4x - 3$$

$$(iv) \quad (f \circ g)(x) = (g \circ f)(x)$$

$$2x^2 + 3 = 4x^2 - 4x + 3$$

$$2x^2 - 4x = 0 \rightarrow x^2 - 2x = 0$$

$$x(x - 2) = 0$$

$$x = 0 \text{ or } x = 2$$

(d)Function : $f : A \rightarrow B$ Domain : A Codomain : B $S, T \subseteq A$

(i) $f(S \cup T) = f(S) \cup f(T)$

Let $y \in f(S \cup T)$ There exists $x \in S \cup T$ [By definition of image of a set such that $f(x) = y$] $x \in S$ or $x \in T$ $f(x) \in f(S)$ or $f(x) \in f(T)$ $f(x) \in f(S) \cup f(T)$ $y \in f(S) \cup f(T)$ $\therefore f(S \cup T) \subseteq f(S) \cup f(T)$ Let $y \in f(S) \cup f(T)$ There exists $x \in S$ or $x \in T$ [By definition of image of a set such that $f(x) = y$] $x \in S \cup T$ $f(x) \in f(S \cup T)$ $y \in f(S \cup T)$ $\therefore f(S) \cup f(T) \subseteq f(S \cup T)$ $\therefore f(S \cup T) = f(S) \cup f(T)$

(ii) $f(S \cap T) \subseteq f(S) \cap f(T)$

Let $y \in f(S \cap T)$ There exists $x \in S \cap T$ [By definition of image of a set such that $f(x) = y$] $x \in S$ and $x \in T$ $f(x) \in f(S)$ and $f(x) \in f(T)$ $f(x) \in f(S) \cap f(T)$ $y \in f(S) \cap f(T)$ $\therefore f(S \cap T) \subseteq f(S) \cap f(T)$

(iii) Let $f : A \rightarrow B$ where $A = \{1, 2\}$ and $B = \{3\}$ such that $f(1) = 3$ and $f(2) = 3$.

Since there are multiple elements of the domain which maps to a single element in codomain, the function f is not one-to-one.

Here $S = \{1\}$ and $T = \{2\}$

$$f(S \cap T) \rightarrow \{1\} \cap \{2\} = \emptyset$$

$$\therefore f(S \cap T) = \emptyset$$

$$f(S) \cap f(T)$$

$$f(\{1\}) = \{3\} \text{ and } f(\{2\}) = \{3\} \rightarrow \{3\} \cap \{3\} = \{3\}$$

$$\therefore f(S) \cap f(T) = \{3\}$$

$$\emptyset \neq \{3\}$$

$$\therefore f(S \cap T) \neq f(S) \cap f(T)$$

(e)

$$(i) \left\lfloor -\frac{17}{4} \right\rfloor = \lfloor -4.25 \rfloor = -5$$

$$(ii) \left\lceil -\frac{17}{4} \right\rceil = \lceil -4.25 \rceil = -4$$

$$(iii) \lfloor \sqrt{20} \rfloor = \lfloor 4.47 \rfloor = 4$$

$$(iv) \lceil \sqrt{20} \rceil = \lceil 4.47 \rceil = 5$$

$$(v) x \in \mathbb{R} \text{ and } n \in \mathbb{Z}$$

$$\lfloor x + n \rfloor \text{ and } \lfloor x \rfloor + n$$

$$\text{Let } m = \lfloor x \rfloor \text{ where } m \leq x < m + 1$$

$$(m + n) \leq x + n < (m + n) + 1 \text{ [Since } m \in \mathbb{Z} \text{ and } n \in \mathbb{Z}, \text{ the sum } (m + n) \in \mathbb{Z}]$$

$$\text{Let } k = \lfloor x + n \rfloor \text{ where } k \leq x + n < k + 1$$

$$\text{From the two inequalities, } m + n = k$$

$$\therefore \lfloor x + n \rfloor = \lfloor x \rfloor + n$$

Question 03

(a)

$$A = \{1, 2, 3, 4, 6, 12\}$$

(i)

$$R = \{ \\ (1, 1), (1, 2), (1, 3), (1, 4), (1, 6), (1, 12), (2, 2), (2, 4), (2, 6), (2, 12), \\ (3, 3), (3, 6), (3, 12), (4, 4), (4, 12), (6, 6), (6, 12), (12, 12) \\ \}$$

(ii) Reflexive :

For the relation to be reflexive, $(a, a) \in R$ and $\forall a \in A$

$(1, 1), (2, 2), (3, 3), (4, 4), (6, 6), (12, 12)$ is in R for all the elements of A.

$\therefore$ R is reflexive.

(iii) Symmetric :

For the relation to be symmetric, $(a, b) \in R \rightarrow (b, a) \in R$.

Here $(1, 2)$ is in R but not $(2, 1)$.

$\therefore$ R is not symmetric.

(iv) Anti-Symmetric :

For the relation to be anti-symmetric, $(a, b) \in R$ and $(b, a) \in R$ implies $a = b$.

Assume $a, b \in A$ where $a|b$ and $b|a$.

Since $a|b$, there exists a positive integer k such that $b = ka$

Since $b|a$, there exists a positive integer m such that $a = mb$

$$a = m(ka) \rightarrow mk = 1$$

Since $m, k \in \mathbb{Z}^+$, $m = k = 1$ which implies $a = b$

$\therefore$ R is anti-symmetric.

(v) Transitive :

For the relation to be transitive, $(a, b) \in R$ and $(b, c) \in R$ implies $(a, c) \in R$.

Assume $a, b, c \in A$ where $a|b$ and $b|c$.

Since $a|b$, there exists a positive integer k such that $b = ka$

Since $b|c$, there exists a positive integer m such that $c = mb$

$$c = m(ka) \rightarrow c = (mk)a \text{ which implies } a|c$$

$\therefore$ R is transitive.

(b)(i) $S = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid 4 \mid (a - b)\}$

(ii) Reflexive :

For the relation to be reflexive, $(a, a) \in S$ and $\forall a \in \mathbb{Z}$ Let $a \in \mathbb{Z}$. (a, a) will always have 0 for $a - a$.Since $0 \in \mathbb{Z}$ and 0 is divisible by 4, aSa always holds for all $a \in \mathbb{Z}$. $\therefore S$ is reflexive.

(iii) Symmetric :

For the relation to be symmetric, $(a, b) \in S \rightarrow (b, a) \in S$.Suppose $a, b \in \mathbb{Z}$ such that aSb where $a - b$ is divisible by 4.Therefore there exists an integer k such that, $a - b = 4k$ Multiplication of both side gives by -1 gives $b - a = -4k \rightarrow b - a = 4(-k)$.Since k is an integer $-k$ is also an integer, therefore bSa holds. $\therefore S$ is symmetric.

(iv) Transitive :

For the relation to be transitive, $(a, b) \in S$ and $(b, c) \in S$ implies $(a, c) \in S$.Assume $a, b, c \in \mathbb{Z}$ where $4 \mid (a - b)$ and $4 \mid (b - c)$.Since $4 \mid (a - b)$, there exists an integer k such that $a - b = 4k$ Since $4 \mid (b - c)$, there exists an integer m such that $b - c = 4m$ $a - b + (b - c) = 4k + 4m \rightarrow a - c = 4(k + m)$ which implies $4 \mid (a - c)$ $\therefore S$ is transitive.

(v) Anti-Symmetric :

For the relation to be anti-symmetric, $(a, b) \in S$ and $(b, a) \in S$ implies $a = b$.Let $a = 5$ and $b = 1$ $a - b = 5 - 1 = 4 = 4(1)$ is divisible by 4. $b - a = 1 - 5 = -4 = 4(-1)$ is divisible by 4.But $5 \neq 1$. $\therefore S$ is not anti-symmetric.

(c)

$$A = \{1, 2, 3, 4\}, B = \{a, b, c\} \text{ and } C = \{x, y, z\}$$

$$\text{Relation of A to B is } R = \{(1, a), (1, b), (2, b), (3, c), (4, a)\}$$

$$\text{Relation of B to C is } S = \{(a, x), (a, z), (b, y), (c, x), (c, z)\}$$

$$(i) \ 1 \in A, a \in B \rightarrow x \text{ and } z = (1, x), (1, z)$$

$$1 \in A, b \in B \rightarrow y = (1, y)$$

$$2 \in A, b \in B \rightarrow y = (2, y)$$

$$3 \in A, c \in B \rightarrow x \text{ and } z = (3, x), (3, z)$$

$$4 \in A, a \in B \rightarrow x \text{ and } z = (4, x), (4, z)$$

$$S \circ R = \{(1, x), (1, y), (1, z), (2, y), (3, x), (3, z), (4, x), (4, z)\}$$

- (ii) For R to be a function, every element of its domain A , must map to exactly one unique element in the codomain B .

In R , 1 maps to both a and b in the codomain.

$\therefore R$ is not a function.

- (iii) For $S \circ R$ to be a function, every element of its domain A , must map to exactly one unique element in the codomain C .

In $S \circ R$, 1 maps to every element in the codomain C .

$\therefore S \circ R$ is not a function.

(d)

$$A = \{1, 2, 3, 4\}$$

$$R = \{(1, 2), (2, 3), (3, 1), (3, 4), (4, 4)\}$$

$$(i) \ 1 \in A, 2 \in A \rightarrow 3 = (1, 3)$$

$$2 \in A, 3 \in A \rightarrow 1 \text{ and } 4 = (2, 1), (2, 4)$$

$$3 \in A, 1 \in A \rightarrow 2 = (3, 2)$$

$$3 \in A, 4 \in A \rightarrow 4 = (3, 4)$$

$$4 \in A, 4 \in A \rightarrow 4 = (4, 4)$$

$$R^2 = R \circ R = \{(1, 3), (2, 1), (2, 4), (3, 2), (3, 4), (4, 4)\}$$

(ii) $1 \in A, 2 \in A \rightarrow 1$ and $4 = (1, 1), (1, 4)$

$$2 \in A, 1 \in A \rightarrow 2 = (2, 2)$$

$$2 \in A, 4 \in A \rightarrow 4 = (2, 4)$$

$$3 \in A, 2 \in A \rightarrow 3 = (3, 3)$$

$$3 \in A, 4 \in A \rightarrow 4 = (3, 4)$$

$$4 \in A, 4 \in A \rightarrow 4 = (4, 4)$$

$$R^3 = R^2 \circ R = \{(1, 1), (1, 4), (2, 2), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

(iii) Relation R is transitive if and only if $R^2 \subseteq R$.

Since $R^2 \not\subseteq R$, R is not transitive.

Pair : (1,3)

$(1, 2) \in R$ and $(2, 3) \in R$, But $(1, 3) \notin R$.

(e)

$$|A| = 4$$

(i) A relation is defined as a subset of cartesian product $A \times A$

$$|A \times A| \rightarrow |A| \cdot |A| \rightarrow |A|^2$$

$$\text{Total number of relations : } 2^{|A|^2} \rightarrow 2^{4^2} \rightarrow 2^{16} = 65536$$

(ii) A relation is reflexive if and only if $(a, a) \in R$ for every $a \in A$

Since there are 4 elements, there are 4 diagonal terms (in the form of (a, a)).

$$\text{Other terms} = |A|^2 - 4 = 16 - 4 = 12$$

$$\text{Total number of reflexive relations : } 2^{12} = 4096$$

(iii) A relation is symmetric if and only if $(a, b) \in R \rightarrow (b, a) \in R$

From 12 non-diagonal terms, $\frac{4(4-1)}{2} = 6$ has symmetric pairs.

$$\text{Total number of symmetric relations : } 2^{4+6} = 2^{10} = 1024$$

Question 04

(a)

$a|b$ means there is an integer k such that $b = ka$ [From definition of divisibility]

$a|c$ means there is an integer m such that $c = ma$ [From definition of divisibility]

$$5b - 3c = 5(ka) - 3(ma) = (5k - 3m)a$$

Since $k, m \in \mathbb{Z}$ let integer $p = (5k - 3m) \in \mathbb{Z}$ by closure of integers.

$$\therefore 5b - 3c = pa$$

$\therefore a|(5b - 3c)$ [From definition of divisibility]

(b)

For contradiction let $\sqrt{3}$ be rational : $\sqrt{3} = \frac{a}{b}$

where $a, b \in \mathbb{Z}, b \neq 0, \gcd(a, b) = 1$ which means a and b don't have common factors

$$3 = \frac{a^2}{b^2} \rightarrow a^2 = 3b^2$$

Since $b^2 \in \mathbb{Z}$, a^2 is a multiple of 3. So $3|a^2$ [From definition of divisibility]

$3|a$ [given the fact if $3|n^2$, then $3|n$]

$3|a$ means there is an integer k such that $a = 3k$ [From definition of divisibility]

$$a^2 = 3b^2 \rightarrow (3k)^2 = 3b^2 \rightarrow 9k^2 = 3b^2 \rightarrow 3k^2 = b^2$$

Since $k^2 \in \mathbb{Z}$, b^2 is a multiple of 3. So $3|b^2$ [From definition of divisibility]

$3|b$ [given the fact if $3|n^2$, then $3|n$]

Since $3|a$ and $3|b$, a and b have a common factor 3.

This contradicts the initial assumption of $\gcd(a, b) = 1$

$\therefore \sqrt{3}$ is irrational.

(c)

(i) $\gcd(414, 662)$

$$1 \quad 662 = 414 * 1 + 248$$

$$2 \quad 414 = 248 * 1 + 166$$

$$3 \quad 248 = 166 * 1 + 82$$

$$4 \quad 166 = 82 * 2 + 2$$

$$5 \quad 82 = 2 * 41 + 0$$

Since the last non-zero remainder is 2, then $\gcd(414, 662) = 2$

$$(ii) \text{ [From 4] : } 166 = 2(82) + 2 \rightarrow 2 = 166 - 2(82)$$

$$\text{[From 3] : } 2 = 166 - 2(248 - 166) \rightarrow 2 = 3(166) - 2(248)$$

$$\text{[From 2] : } 2 = 3(414 - 248) - 2(248) \rightarrow 2 = 3(414) - 5(248)$$

$$\text{[From 1] : } 2 = 3(414) - 5(662 - 414) \rightarrow 2 = 8(414) - 5(662)$$

$$2 = 414(8) + 662(-5)$$

$$2 = 414s + 662t \text{ [Here } s = 8 \text{ and } t = -5 \text{]}$$

(d)

$$18x \equiv 30 \pmod{42}$$

To determine the number of solutions $\gcd(18, 42)$

$$18 = 2 \cdot 3^2 \text{ and } 42 = 2 \cdot 3 \cdot 7$$

$$\gcd = 2 \cdot 3 = 6$$

$\therefore$ there are 6 incongruent solutions.

$$\frac{18}{6}x \equiv \frac{30}{6} \pmod{\frac{42}{6}} \rightarrow 3x \equiv 5 \pmod{7}$$

$$\gcd(7, 3)$$

$$1 \quad 7 = 3 \cdot 2 + 1$$

$$2 \quad 3 = 1 \cdot 3 + 0$$

$$\text{[From 1] : } 1 = 7 - 2(3) \rightarrow 1 = 3(-2) + 7$$

$$1 \equiv -2(3) \pmod{7}$$

Since $-2 \equiv 5 \pmod{7}$, modular inverse of $3 \pmod{7}$ is 5

$$3x \equiv 5 \pmod{7} \rightarrow x \equiv 5 \times 5 \pmod{7} \rightarrow x \equiv 25 \pmod{7} \rightarrow x \equiv 4 \pmod{7}$$

For incongruent solutions, $x_k = 4 + 7 \cdot k$ where $k = \{0, 1, 2, 3, 4, 5\}$

$$k = 0 \rightarrow x_0 = 4 + 7(0) = 4$$

$$k = 1 \rightarrow x_1 = 4 + 7(1) = 11$$

$$k = 2 \rightarrow x_2 = 4 + 7(2) = 18$$

$$k = 3 \rightarrow x_3 = 4 + 7(3) = 25$$

$$k = 4 \rightarrow x_4 = 4 + 7(4) = 32$$

$$k = 5 \rightarrow x_5 = 4 + 7(5) = 39$$

Set of incongruent solutions is $\{4, 11, 18, 25, 32, 39\}$

Verification for $18x \equiv 30 \pmod{42}$

$$x = 4 \rightarrow 18(4) = 72 = 42 + 30 = 30 \pmod{42}$$

$$x = 11 \rightarrow 18(11) = 198 = 4(42) + 30 = 30 \pmod{42}$$

$$x = 18 \rightarrow 18(18) = 324 = 7(42) + 30 = 30 \pmod{42}$$

$$x = 25 \rightarrow 18(25) = 450 = 10(42) + 30 = 30 \pmod{42}$$

$$x = 32 \rightarrow 18(32) = 576 = 13(42) + 30 = 30 \pmod{42}$$

$$x = 39 \rightarrow 18(39) = 702 = 16(42) + 30 = 30 \pmod{42}$$

(e)

$$P(n) : p|a^n \rightarrow p|a$$

$$[n = 1]P(1) : p|a^1 \rightarrow p|a$$

$\therefore$ for $n = 1$, $P(n)$ is true for all integers of a .

$$\text{Assume } P(k) : p|a^k \rightarrow p|a$$

$$P(k+1) : p|a^{k+1} \rightarrow p|a$$

$$P(k+1) : p|(a^k \cdot a) \rightarrow p|a$$

$$p|(a^k \cdot a) = p|a^k \text{ or } p|a$$

$p|a^k$ is true from the assumption.

$p|a$ is the conclusion directly.

$\therefore$ for $P(k+1)$ the conclusion holds.

By principle of mathematical induction $p|a^n \rightarrow p|a$

For contradiction let $\sqrt{p}$ be rational : $\sqrt{p} = \frac{a}{b}$

where $a, b \in \mathbb{Z}, b \neq 0, \gcd(a, b) = 1$ which means a and b don't have common factors

$$p = \frac{a^2}{b^2} \rightarrow a^2 = pb^2$$

Since $b^2 \in \mathbb{Z}$, a^2 is a multiple of p . So $p|a^2$

$p|a$

$p|a$ means there is an integer k such that $a = pk$

$$a^2 = pb^2 \rightarrow (pk)^2 = pb^2 \rightarrow p^2k^2 = pb^2 \rightarrow pk^2 = b^2$$

Since $k^2 \in \mathbb{Z}$, b^2 is a multiple of p . So $p|b^2$

$p|b$

Since $p|a$ and $p|b$, a and b have a common factor p .

This contradicts the initial assumption of $\gcd(a, b) = 1$

$\therefore \sqrt{p}$ is irrational.

[From definition of divisibility]

[proved above $p|a^n$, then $p|a$]

[From definition of divisibility]

[From definition of divisibility]

[proved above $p|a^n$, then $p|a$]